

Exploring USB TYPE-C And USB POWER DELIVERY

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Every now and then there is an innovation or advancement that changes the way we operate, the way the world operates. These innovations and advancements in technology create opportunities for businesses and provide improved features to consumers. These are also widely and swiftly adopted. USB Type-C is one such technology advancement that will soon be found in most devices. Apple and Google have taken the lead, whereas companies like Dell and many more are following suit.

USB interface is the most widely-adopted interface specification that is being used across all industries. This interface has slowly eliminated legacy serial and parallel ports. Now with advancements in computing technologies, devices are far more efficient and have reduced form factors. This has necessitated a change in the existing USB connectors.

USB Type-C specification is targeted towards new, compact and power-hungry devices and brings in a new flavour to the USB connector. This new ecosystem is also managed by another new specification, namely, USB Power Delivery (PD) specification, which allows managing power and functionality

over the new USB Type-C connector.

This article aims to provide an insight into USB Type-C specification, USB PD specification and new functionalities you shall see and experience with the new USB Type-C ecosystem.

USB Type-C overview

USB Type-C cable and connector specifications cover mechanical, assembly and connection detection and configuration part of the USB Type-C ecosystem.

Unlike the legacy USB connector that has separate connectors for host and devices, Type-C connector uses the same connector for both host and device. The host and device relationship is negotiated using the configuration channel (CC) signal.

USB Type-C interface can be classified into two major types, namely, USB Type-C receptacle and USB Type-C plug.

USB Type-C receptacle. USB Type-C interface on a host or device is generally a receptacle connector as illustrated in Fig. 1.

The receptacle in Fig. 1 provides multiple sets of signals to enable the desired functionality irrespective of the plug orientation. For a full-featured USB Type-C receptacle, all indicated signals are required.

USB Type-C plug. USB Type-C plug interface is a connector that is present in a cable or device like mouse or flash drive with only one CC pin and one set of USB 2.0 lines as shown in Fig. 2. The single CC pin allows orientation detection and the other CC pin of the receptacle is used for sourcing V_{CONN} .

It is important to note that the super-speed lines are cross-connected on Tx and Rx in a cable.

Let us now take a brief look at the signal lines on the USB Type-C connector.

A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12
GND	TX1+	TX-	V _{US}	CC1	D+	D-	SBU1	V _{US}	RX2-	RX2+	GND
GND	RX1+	RX1-	V _{US}	SBU2	D-	D+	CC2	V _{US}	TX2-	TX2+	GND
B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1

Fig. 1: USB Type-C receptacle (Ref: USB Type-C specification)

A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1
GND	RX2+	RX2-	V _{US}	SBU1	D-	D+	CC	V _{US}	TX1-	TX1+	GND
GND	TX2+	TX2-	V _{US}	V _{CONN}			SBU2	V _{US}	RX1-	RX1+	GND
B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12

Fig. 2: USB Type-C plug (Ref: USB Type-C specification)